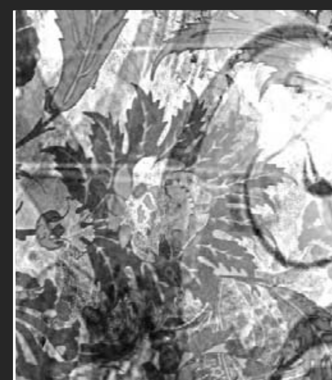
THE SCIENTIFIC MOVEMENTS LEADING TO EVOLUTIONARY PSYCHOLOGY



From Chapter 1 of *Evolutionary Psychology: The New Science of the Mind,* Fourth Edition. David M. Buss.

In the distant future I see open fields for more important researches. Psychology will be based on a new foundation, that of the necessary acquirement of each mental power and capacity by gradation.

—Charles Darwin, 1859

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As the archeologist dusted off the dirt and debris from the skeleton, she noticed something strange: The left side of the skull had a large dent, apparently from a ferocious blow, and the rib cage—also on the left side—had the head of a spear lodged in it. Back in the laboratory, scientists determined that the skeleton was that of a Neanderthal man who had died roughly 50,000 years ago, the earliest known homicide victim. His killer, judging from the damage to the skull and rib cage, bore the lethal weapon in his right hand.

The fossil record of injuries to bones reveals two strikingly common patterns (Jurmain et al., 2009; Trinkaus & Zimmerman, 1982; Walker, 1995). First, the skeletons of men contain far more fractures and dents than do the skeletons of women. Second, the injuries are located mainly on the left frontal sides of the skulls and skeletons, suggesting right-handed attackers. The bone record alone cannot tell us with certainty that combat among men was a central feature of human ancestral life. Nor can it tell us with certainty that men evolved to be the more physically aggressive sex. But skeletal remains provide clues that yield a fascinating piece of the puzzle of where we came from, the forces that shaped who we are, and the nature of our minds today.

The huge human brain, approximately 1,350 cubic centimeters, is the most complex organic structure in the known world. Understanding the human mind/brain mechanisms in evolutionary perspective is the goal of the new scientific discipline called *evolutionary psychology*. Evolutionary psychology focuses on four key questions: (1) *Why* is the mind designed the way it is—that is, what causal processes created, fashioned, or shaped the human mind into its current form? (2) *How* is the human mind designed—what are its mechanisms or component parts, and how are they organized? (3) *What are the functions* of the component parts and their organized structure—that is, what is the mind designed to do? (4) *How* does input from the current environment interact with the design of the human mind to produce observable behavior?

Contemplating the mysteries of the human mind is not new. Ancient Greeks such as Aristotle and Plato wrote manifestos on the subject. More recently, theories of the human mind such as the Freudian theory of psychoanalysis, the Skinnerian theory of reinforcement, and connectionism have vied for the attention of psychologists.

Only within the past few decades have we acquired the conceptual tools to synthesize our understanding of the human mind under one unifying theoretical framework—that of evolutionary psychology. This discipline pulls together findings from all disciplines of the mind, including those of brain imaging; learning and memory; attention, emotion, and passion; attraction, jealousy, and sex; self-esteem, status, and self-sacrifice; parenting, persuasion, and perception; kinship, warfare, and aggression; cooperation, altruism, and helping; ethics, morality, and medicine; and commitment, culture, and consciousness. This book offers an introduction to evolutionary psychology and provides a road map to this new science of the mind.

This chapter starts by tracing the major landmarks in the history of evolutionary biology that were critical in the emergence of evolutionary psychology. Then we turn to the history of the field of psychology and show the progression of accomplishments that led to the need for integrating evolutionary theory with modern psychology.

■ LANDMARKS IN THE HISTORY OF EVOLUTIONARY THINKING

We begin our examination of the history of evolutionary thinking well before the contributions of Charles Darwin and then consider the various milestones in its development through the end of the twentieth century.

Evolution before Darwin

Evolution refers to change over time. Change in life forms was postulated by scientists to have occurred long before Darwin published his classic 1859 book, *On the Origin of Species* (see Glass, Temekin, & Straus, 1959; and Harris, 1992, for historical treatments).

Jean Baptiste Pierre Antoine de Monet Chevalier de Lamarck (1744–1829) was one of the first scientists to use the word *biologie*, thus recognizing the study of life as a distinct science. Lamarck believed in two major causes of species change: first, a natural tendency for each species to progress toward a higher form and, second, the inheritance of acquired characteristics. Lamarck said that animals must struggle to survive and this struggle causes their nerves to secrete a fluid that enlarges the organs involved in the struggle. Giraffes evolved long necks, he thought, through their attempts to eat from higher and higher leaves (recent evidence suggests that long necks may also play a role in mate competition). Lamarck believed that the neck changes that came about from these strivings were passed down to succeeding generations of giraffes, hence the phrase “the inheritance of acquired characteristics.” Another theory of change in life forms was developed by Baron Georges Léopold Chrétien Frédérick Dagobert Cuvier

(1769–1832). Cuvier proposed a theory called *catastrophism*, according to which species are extinguished periodically by sudden catastrophes, such as meteorites, and then replaced by different species.

Biologists before Darwin also noticed the bewildering variety of species, some with astonishing structural similarities. Humans, chimpanzees, and orangutans, for example, all have exactly five digits on each hand and foot. The wings of birds are similar to the flippers of seals, perhaps suggesting that one was modified from the other (Daly & Wilson, 1983). Comparisons among these species suggested that life was not static, as some scientists and theologians had argued. Further evidence suggesting change over time also came from the fossil record. Bones from older geological strata were not the same as bones from more recent geological strata. These bones would not be different, scientists reasoned, unless there had been a change in organic structure over time.

Another source of evidence came from comparing the embryological development of different species (Mayr, 1982). Biologists noticed that such development was strikingly similar in species that otherwise seemed very different from one another. An unusual loop-like pattern of arteries close to the bronchial slits characterizes the embryos of mammals, birds, and frogs. This evidence suggested, perhaps, that these species might have come from the same ancestors millions of years ago. All these pieces of evidence, present before 1859, suggested that life was not fixed or unchanging. The biologists who believed that organic structure changed over time called themselves evolutionists.

Another key observation had been made by various evolutionists before Darwin: Many species possess characteristics that seem to have a purpose. The porcupine's quills help it fend off predators. The turtle's shell helps to protect its tender organs from the hostile forces of nature. The beaks of many birds are designed to aid in cracking nuts. This apparent functionality, so abundant in nature, required an explanation.

Missing from the evolutionists' accounts before Darwin, however, was a theory to explain how change might take place over time and how such seemingly purposeful structures such as the giraffe's long neck and the porcupine's sharp quills could have come about. A causal mechanism or process to explain these biological phenomena was needed. Charles Darwin provided the theory of just such a mechanism.

Darwin's Theory of Natural Selection

Darwin's task was more difficult than it might at first appear. He wanted not only to explain why change takes place over time in life forms, but also to account for the particular ways it proceeds. He wanted to determine how new species emerge (hence the title of his book *On the Origin of Species*), as well as how others vanish. Darwin wanted to explain why the component parts of animals—the long necks of giraffes, the wings of birds, and the trunks of elephants—existed in those particular forms. And he wanted to explain the apparent purposive quality of those forms, or why they seem to function to help organisms accomplish specific tasks.

The answers to these puzzles can be traced to a voyage Darwin took after graduating from Cambridge University. He traveled the world as a naturalist on a ship, the *Beagle*, for a five-year period, from 1831 to 1836. During this voyage, he collected dozens of samples of birds and other animals from the Galápagos Islands in the Pacific Ocean. On returning from his voyage, he discovered that the Galápagos finches, which he had presumed were all of the same species,

North Wind Picture Archives

Charles Darwin created a scientific revolution in biology with his theory of natural selection. His book On the Origin of Species *(1859) is packed with theoretical arguments and detailed empirical data that he amassed over the twenty-five years prior to the book's publication.*

actually varied so much that they constituted different species. Indeed, each island in the Galápagos had a distinct species of finch. Darwin determined that these different finches had a common ancestor but had diverged from each other because of the local ecological conditions on each island. This geographic variation was pivotal to Darwin's conclusion that species are not immutable but can change over time.

What could account for why species change? Darwin struggled with several different theories of the origins of change, but rejected all of them because they failed to explain a critical fact: the existence of adaptations. Darwin wanted to account for change, of course, but he also wanted to account for why organisms appeared so well designed for their local environments.

> It was . . . evident that [these other theories] could [not] account for the innumerable cases in which organisms of every kind are beautifully adapted to their habits of life—for instance, a woodpecker or tree-frog to climb trees, or a seed for dispersal by hooks and plumes. I had always been much struck by such adaptations, and until these could be explained it seemed to me almost useless to endeavour to prove by indirect evidence that species have been modified. (Darwin, from his autobiography; cited in Ridley, 1996, p. 9)

Darwin unearthed a key to the puzzle of adaptations in Thomas Malthus's *An Essay on the Principle of Population* (published in 1798), which introduced Darwin to the notion that organisms exist in numbers far greater than can survive and reproduce. The result must be a "struggle for existence," in which favorable variations tend to be preserved and unfavorable ones tend to die out. When this process is repeated generation after generation, the end result is the formation of new adaptation.

More formally, Darwin's answer to all these puzzles of life was the theory of *natural selection* and its three essential ingredients: *variation, inheritance,* and *selection.*[1] First, organisms vary in all sorts of ways, such as in wing length, trunk strength, bone mass, cell structure, fighting ability, defensive ability, and social cunning. Variation is essential for the process of evolution to operate—it provides the "raw materials" for evolution.

Second, only some of these variations are inherited—that is, passed down reliably from parents to their offspring, which then pass them on to their offspring down through the generations. Other variations, such as a wing deformity caused by an environmental accident, are not inherited by offspring. Only those variations that are inherited play a role in the evolutionary process.

The third critical ingredient of Darwin's theory is selection. Organisms with some heritable variants leave more offspring *because* those attributes help with the tasks of *survival* or

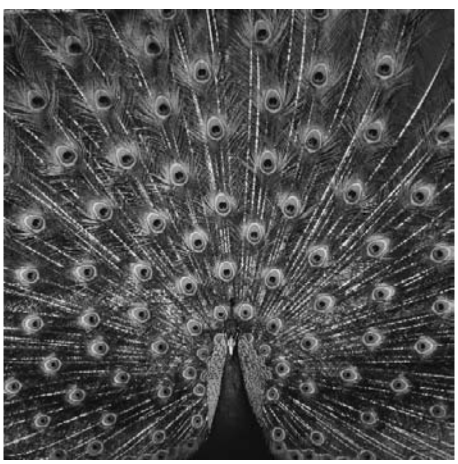
[1]The theory of natural selection was discovered independently by Alfred Russel Wallace (Wallace, 1858); Darwin and Wallace co-presented the theory at a meeting of the Linnaen Society.

reproduction. In an environment in which the primary food source might be nut-bearing trees or bushes, some finches with a particular shape of beak, for example, might be better able to crack nuts and get at their meat than finches with other shapes of beaks. More finches who have beaks better shaped for nut cracking survive than those with beaks poorly shaped for nut cracking.

An organism can survive for many years, however, and still not pass on its inherited qualities to future generations. To pass its inherited qualities to future generations, it must reproduce. Thus, *differential reproductive success*, brought about by the possession of heritable variants that increase or decrease an individual's chances of surviving and reproducing, is the "bottom line" of evolution by natural selection. Differential reproductive success or failure is defined by reproductive success relative to others. The characteristics of organisms that reproduce more than others, therefore, get passed down to future generations at a relatively greater frequency. Because survival is usually necessary for reproduction, it took on a critical role in Darwin's theory of natural selection.

Darwin's Theory of Sexual Selection

Darwin had a wonderful scientific habit of noticing facts that seemed inconsistent with his theories. He observed several that seemed to contradict his theory of natural selection, also called "survival selection." First he noticed weird structures that seemed to have absolutely nothing to do with survival; the brilliant plumage of peacocks was a prime example. How could this strange luminescent structure possibly have evolved? The plumage is obviously metabolically costly to the peacock. Furthermore, it seems like an open invitation to predators. Darwin became so obsessed with this apparent anomaly that he once commented, "The sight of a feather in a peacock's tail, whenever I gaze at it makes me sick!" (quoted in Cronin, 1991, p. 113). Darwin also observed that in some species, the sexes differed dramatically in size and structure. Why would the sexes differ so much, Darwin pondered, when both have essentially the same problems of survival, such as eating, fending off predators, and combating diseases?



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Darwin got sick at the sight of a peacock because, initially, the brilliant plumage seemed to have no obvious survival value and hence could not be explained by his original theory of natural selection. He eventually developed the theory of sexual selection, which could explain the peacock's plumage, and presumably he stopped getting sick when he witnessed one.

Darwin's answer to these apparent embarrassments to the theory of natural selection was to devise a second evolutionary theory: the theory of *sexual selection.* In contrast to the theory of natural selection, which focused on adaptations that have arisen as a consequence of successful survival, the theory of sexual selection focused on adaptations that arose as a consequence of successful mating. Darwin envisioned two primary means by which sexual selection could operate. The first is *intrasexual competition*—competition between members of one sex, the outcomes of which contributed to mating access to the other sex. The prototype of intrasexual competition is two stags locking horns in combat. The victor gains

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Stags locking horns in combat is a form of sexual selection called intrasexual competition. The qualities that lead to success in these same-sex combats get passed on in greater numbers to succeeding generations because the victors gain increased mating access to members of the opposite sex.

sexual access to a female either directly or through controlling territory or resources desired by the female. The loser typically fails to mate. Whatever qualities lead to success in the same-sex contests, such as greater size, strength, or athletic ability, will be passed on to the next generation by virtue of the mating success of the victors. Qualities that are linked with losing fail to get passed on. So evolution—change over time—can occur simply as a consequence of intrasexual competition.

The second means by which sexual selection could operate is *intersexual selection*, or preferential mate choice. If members of one sex have some consensus about the qualities that are desired in members of the opposite sex, then individuals of the opposite sex who possess those qualities will be preferentially chosen as mates. Those who lack the desired qualities fail to get mates. In this case, evolutionary change occurs simply because the qualities that are desired in a mate increase in frequency with the passing of each generation. If females prefer to mate with males who give them nuptial gifts, for example, then males with qualities that lead to success in acquiring nuptial gifts will increase in frequency over time. Darwin called the process of intersexual selection *female choice* because he observed that throughout the animal world, females of many species were discriminating or choosy about whom they mated with.

Darwin's theory of sexual selection succeeded in explaining the anomalies that worried him. The peacock's tail, for example, evolved because of the process of intersexual selection: Peahens prefer to mate with males who have the most brilliant and luminescent plumage. Males are often larger than females in species in which males engage in physical combat with other males for sexual access to females—the process of intrasexual competition.

The Role of Natural Selection and Sexual Selection in Evolutionary Theory

Darwin's theories of natural and sexual selection are relatively simple to describe, but many sources of confusion surround them even to this day. This section clarifies some important aspects of selection and its place in understanding evolution.